

Assignment – Week 7

Introduction to Machine Learning-IIT KGP

Topic: Computational Learning Theory and Ensemble Learning

TYPE OF QUESTION: MCQ/MSQ

Number of Questions: 10

Total Marks: $10 \times 2 = 20$

Question 1

For a particular learning task, if the required error parameter ϵ changes from 0.1 to 0.01, then how many more samples will be required for PAC learning?

- A. Same
- B. 2 times
- C. 10 times
- D. 1000 times

Correct Answer : C. 10 times

Detailed Solution:

$$m \geq (1/\epsilon)(\log(|H|) + \log(1/\delta))$$

Where m is the number of samples, ϵ is the error parameter, H is the hypothesis space and δ is the confidence parameter.

m is inversely proportional to the error parameter ϵ .

$$m_1/m_2 = \epsilon_2/\epsilon_1$$

$$m_2 = m_1 \cdot (\epsilon_1/\epsilon_2)$$

$$m_2 = m_1 \cdot (0.1/0.01)$$

$$m_2 = m_1 \cdot 10$$

Question 2

Suppose a hypothesis space has VC dimension 8. Which statement is necessarily true?

- A) Every set of 8 points can be shattered.
- B) There exists at least one set of 8 points that can be shattered.
- C) Some set of 9 points can be shattered.
- D) Every set of 9 points can be shattered.

Correct Answer: B

Detailed Solution:

A VC dimension of 8 means there exists at least one set of 8 points that can be

shattered by the hypothesis class. It does not imply that every set of 8 points can be shattered. Furthermore, no set of 9 points can be shattered.

Question 3

Which best describes the purpose of bootstrap sampling in bagging?

- A) Increase bias
- B) Create diverse training datasets
- C) Reduce training examples
- D) Ensure identical data

Correct Answer: B

Detailed Solution: Bagging creates multiple bootstrap samples by randomly sampling the training data with replacement. Different training sets lead to diverse base learners, which reduces variance and improves generalization.

Question 4

Which statements about PAC learning are correct?

1. PAC stands for Probably Approximately Correct.
2. PAC learning provides generalization guarantees.
3. PAC learning uses confidence and accuracy parameters.

- A) 1 only
- B) 2 only
- C) 1 and 2 only
- D) 1, 2 and 3

Correct Answer: D

Detailed Solution:

PAC stands for Probably Approximately Correct. The framework provides guarantees on the generalization error of a learned hypothesis using the parameters ϵ (accuracy) and δ (confidence).

Question 5

A learning algorithm achieves 98% training accuracy but only 70% test accuracy. What is most likely occurring?

- A) Underfitting
- B) Overfitting
- C) Bagging
- D) Boosting

Correct Answer: B

Detailed Solution: The model performs extremely well on the training data but poorly on unseen data. This indicates that it has memorized training patterns and noise instead of learning generalizable relationships.

Question 6

Which can help reduce overfitting?

1. Increasing training data
2. Bagging
3. Regularization
4. Increasing model complexity without constraint

- A) 1 and 2 only
- B) 1, 2 and 3 only
- C) 2 and 4 only
- D) 1, 3 and 4 only

Correct Answer: B

Detailed Solution:

More data provides better estimates of the underlying distribution. Bagging reduces variance, and regularization discourages overly complex models. Increasing complexity without constraint usually worsens overfitting.

Question 7

Suppose an AdaBoost ensemble consists of three weak classifiers h_1 , h_2 , and h_3 . After training, the corresponding classifier weights are: $\alpha_1 = 0.6$, $\alpha_2 = 0.4$, $\alpha_3 = 0.3$.

For a particular test instance x , the predictions of the three weak classifiers are:

$h_1(x) = 1$, $h_2(x) = -1$, $h_3(x) = 1$. Using the AdaBoost decision rule, what will be the final prediction of the ensemble classifier $H(x)$?

- A) +1
- B) -1
- C) 0
- D) Cannot be determined

Correct Answer: A) +1

Detailed Solution: In AdaBoost, the final prediction is obtained by taking the sign of the weighted sum of the predictions made by all weak classifiers.

The ensemble prediction is given by:

$$H(x) = \text{sign}(\alpha_1 h_1(x) + \alpha_2 h_2(x) + \alpha_3 h_3(x))$$

Substituting the given values: $H(x) = \text{sign}(0.6 \times 1 + 0.4 \times (-1) + 0.3 \times 1)$

$= \text{sign}(0.6 - 0.4 + 0.3) = \text{sign}(0.5)$ Since $0.5 > 0$, we have: $H(x) = +1$

Question 8

A weak learner in the Adaboost algorithm has weighted error $\epsilon=0.2$. Compute the classifier weight α (alpha).

- A) 0.35
- B) 0.69
- C) 1.10
- D) 1.39

Correct Answer: B

Detailed Solution:

$\alpha = 0.5 \ln(0.8/0.2) = 0.5 \ln(4) \approx 0.5 \times 1.386 = 0.693$. Hence the closest option is 0.69.

Question 9

In AdaBoost, we give more weights to points having been misclassified in previous iterations. Now, if we introduce a limit or cap on the weight that any point can take (for example, say we introduce a restriction that prevents any point's weight from exceeding a value of 10). Which among the following would be the effect of such a modification? (Select ALL that apply)

- A) It will have no effect on the performance of the Adaboost method.
- B) It makes the final classifier robust to outliers.
- C) It may result in lower overall performance.
- D) No effect.

Correct Answer: B, C

Detailed Solution: Outliers tend to get misclassified. As the number of iterations increases, the weight corresponding to outlier points can become very large resulting in subsequent classifier models trying to classify the outlier points correctly. This generally has an adverse effect on the overall classifier. Restricting the weights is one way of mitigating this problem. However, this can also lower the performance of the classifier.

Question 10

Identify whether the following statement is true or false:

"Boosting is easy to parallelize whereas bagging is inherently a sequential process."

- A) True
- B) False

Correct Answer: B) False.

Detailed Solution: Bagging is easy to parallelize whereas boosting is inherently a sequential process.
